Their hybrid shows the presence of both the types of proteins similar to their parents.

The heterozygote genotype gives rise to a phenotype distinctly different from either of the homozygous genotypes. The F_1 heterozygotes produce a F_2 progeny in a phenotypic and genotypic ratios of 1 : 2 : 1.

2.4.3. Lethal genes

An allele which has the potential to cause the death of an organism is called a "Lethal Allele". In 1907, E. Baur reported a lethal gene in snapdragon (*Antirrhinum sp.*). It is an example for recessive lethality. In snapdragon there are three kinds of plants.

1. Green plants with chlorophyll. (CC)
2. Yellowish green plants with carotenoids are referred to as pale green, golden or aurea plants (Cc)
3. White plants without any chlorophyll. (cc)

The genotype of the homozygous green plants is CC. The genotype of the homozygous white plant is cc.

The aurea plants have the genotype Cc because they are heterozygous of green and white plants. When two such aurea plants are crossed the F_1 progeny has identical phenotypic and genotypic ratio of 1 : 2 : 1 (viz. 1 Green (CC) : 2 Aurea (Cc) : 1 White (cc))

Since the white plants lack chlorophyll pigment, they will not survive. So the F_2 ratio is modified into 1 : 2. In this case the homozygous recessive genotype (cc) is lethal.

F_1	Heterozygote	Antirrhinum aurea	x	Antirrhinum aurea		
		Cc	x	Cc		
F_2		1CC	:	2 Cc	:	1 cc
		Green		Aurea		White (lethal)

Figure: 2.15: Lethal genes

The term "lethal" is applied to those changes in the genome of an organism which produces effects severe enough to cause death. Lethality is a condition in which the death of certain genotype occurs prematurely. The fully dominant or fully recessive lethal allele kills the carrier individual only in its homozygous condition. So the F_2 genotypic ratio will be 2 : 1 or 1 : 2 respectively.

2.4.4. Pleiotropy – A single gene affects multiple traits

In Pleiotropy, the single gene affects multiple traits and alter the phenotype of the organism. The Pleiotropic gene influences a number of characters simultaneously and such genes are called pleiotropic gene. These were crossed with a variety of peas having white flowers, light coloured seeds and no spot on the axils of the leaves, the three traits for flower colour, seed colour and a leaf axil spot all were inherited together as a single unit. Another example is sickle cell anemia.

2.5 Intergenic gene interactions



Interlocus interactions take place between the alleles at different loci i.e between alleles of different genes.It includes the following:

Dominant Epistasis – It is a gene interaction in which two alleles of a gene at one locus interfere and suppress or mask the phenotypic expression of a different pair of alleles of another gene at another locus. The gene that suppresses or masks the phenotypic expression of a gene at another locus is known as **epistatic**. The gene whose expression is interfered by non-allelic genes and prevents from exhibiting its character is known as **hypostatic**. When both the genes are present together, the phenotype is determined by the epistatic gene and not by the hypostatic gene.

In the summer squash the fruit colour locus has a dominant allele 'W' for white

TN_GOVT_BOTANY_XII_PAGES_035-054 CH 02.indd 44 3/8/2020 2:42:39 PM